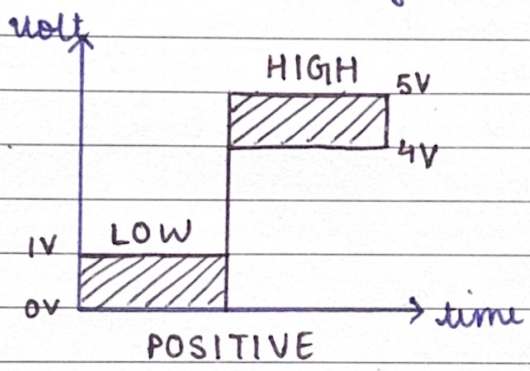
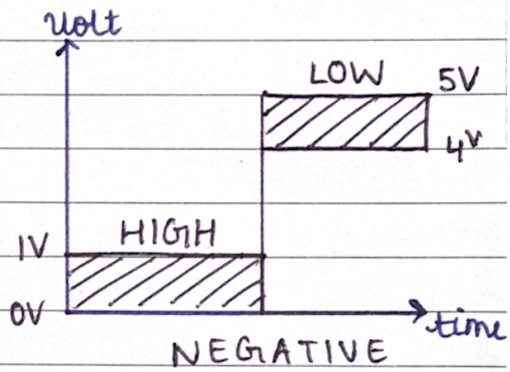
Topic _____ Date _____

→ Digital Logic Families

- A group of compatible IC's with same logic levels and supply voltages fabricated for performing various logical functions is refurred as logic family.
- There are two types of logic levels

1. Positive Logic : Logic is high at high voltage and low on low voltage
2. Negative Logic : Logic is high at low voltage and low at high voltage.

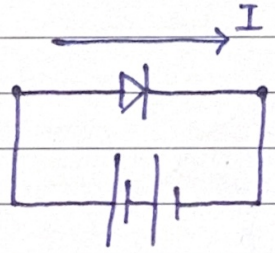




→ Semiconductor Devices

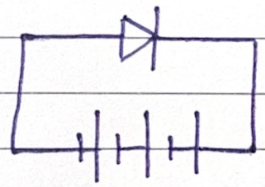
There are two types of semiconductor devices

1. Bipolar : It has two charge carriers i.e. electrons & holes
2. Unipolar: It has only one charge carrier i.e either electron or holes



Diode (ON)

Forward Bias



No I flows in this case

Diode (OFF)

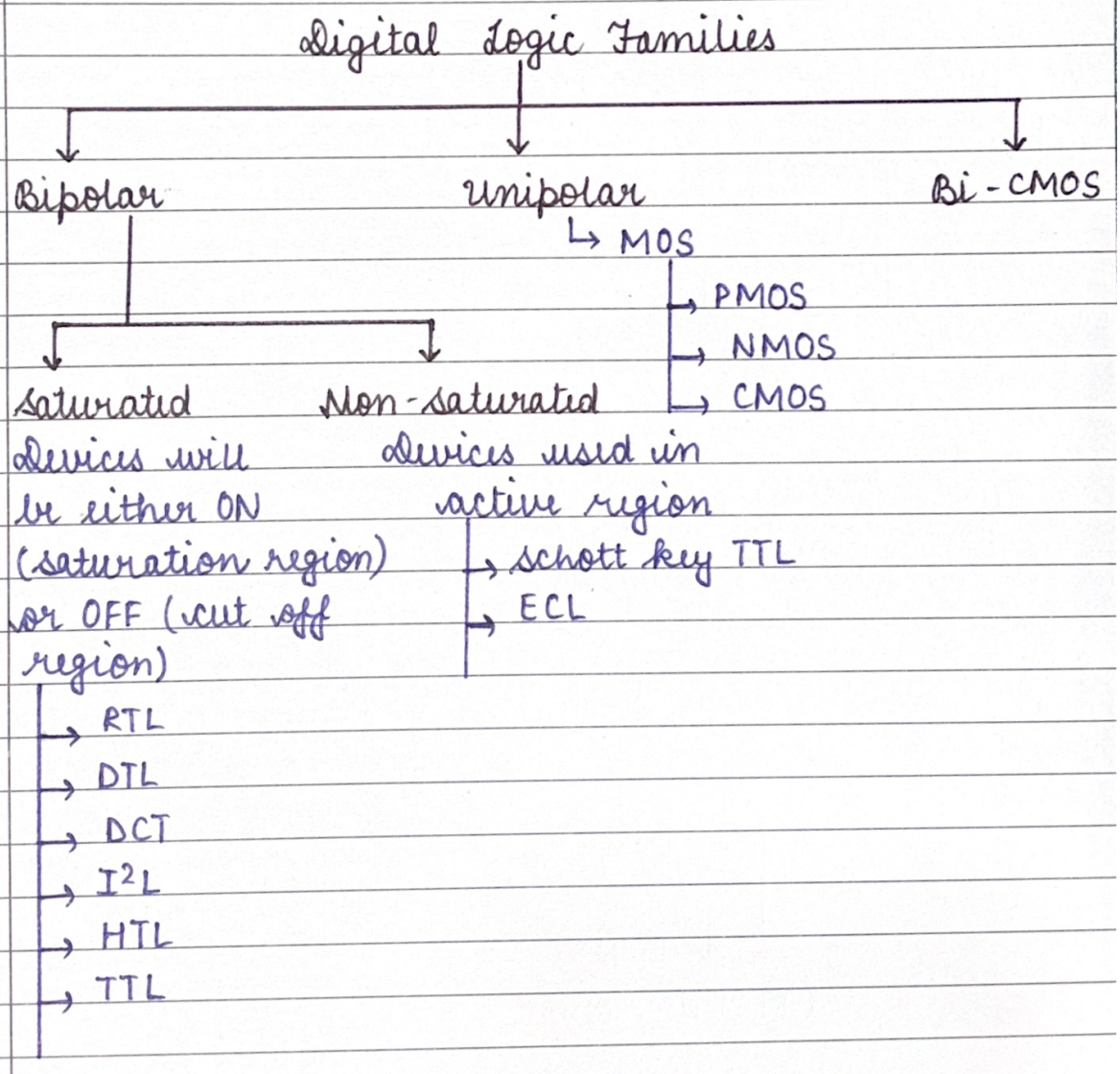
Reverse Bias

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Topic _____ Date _____

→ Classification of Digital Logic Families

1. Bipolar logic families
   - Uses electrons & holes as charge carriers
2. Unipolar logic families
   - Uses either e^- or holes as charge carrier
3. Bi-CMOS logic families
   - It uses MOS that's bipolar
   - Input & logical operations performed by CMOS whereas output performed by MOS



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→ Merits of digital IC
- Reduces overall size of system
- Reduces overall cost of system
- Improves reliability of system by removing inter connection of wiring
- Reduces power consumption of system

→ Demerits of digital IC
- It can't handle large power
- Electrical Devices like resistor, transistor, inductor & large capacitors can't be implemented with great precision on chip
- Only suitable for low power applications

→ Characterstics of digital IC

1. Propagation Delay : It justifies spud of the operation of digital IC and explains the amount of delay happened when signal is transitioning from input to output

2. Thresold voltage : It explains the amount of voltage at which transition occurs from logic low to logic high

3. Power Dissipation : It justifies the amount of power dissipated on IC in form of heat

4. Figure of merit : It's the product of the propagation delay & power dissipation

5. Fan out & Fan In : Fan out explains how many similar gates can be driven by

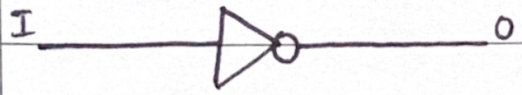
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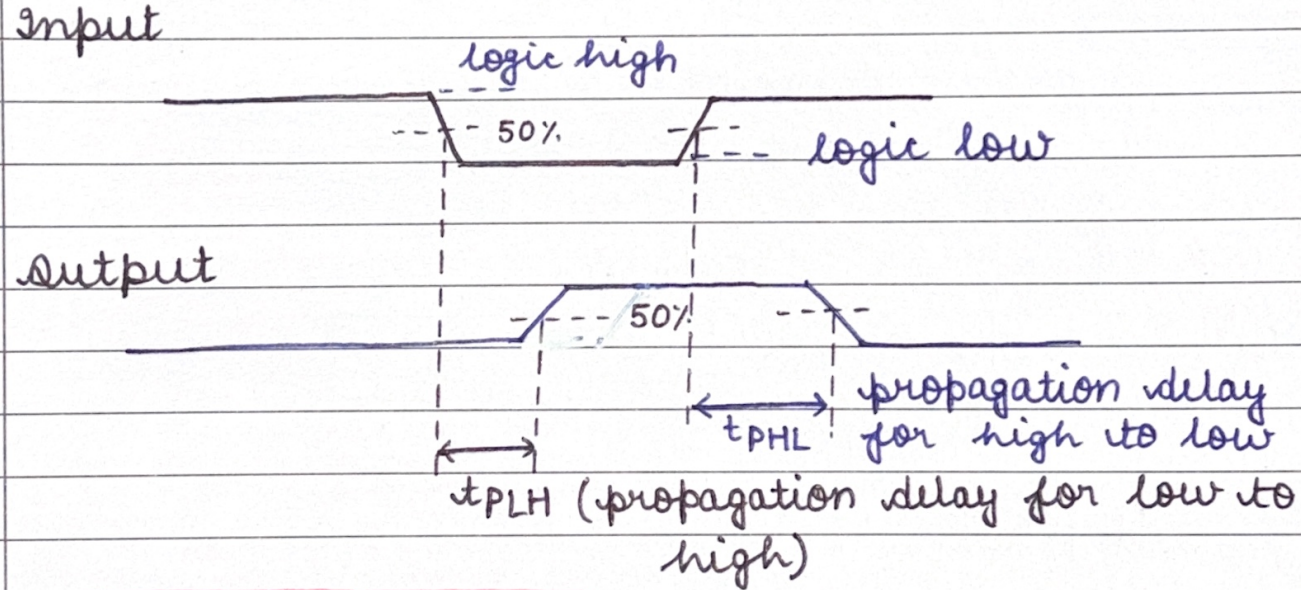
Topic _____ Date _____

given logic gate. Fan in explains no. of inputs that can be handled at a given logical gate.

6. Noise Immunity

7. Operating Time





$$t_p = \frac{t_{PLH} + t_{PHL}}{2} \rightarrow \text{Propagation Delay}$$

$P_d = V_{cc} \times I_{cc}(avg)$ where $I_{CC} = \frac{I_{CCH} + I_{CCL}}{2}$

↳ V_{cc} in TTL & V_{DD} in CMOS

I_{CCH} → I_{cc} at logic high

I_{CCL} → I_{cc} at logic low

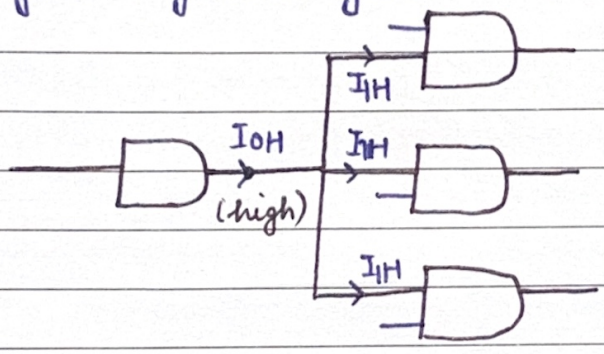
$$\Rightarrow P_d = V_{cc} \cdot \left[\frac{I_{CCH} + I_{CCL}}{2}\right] \rightarrow \text{Power dissipation}$$

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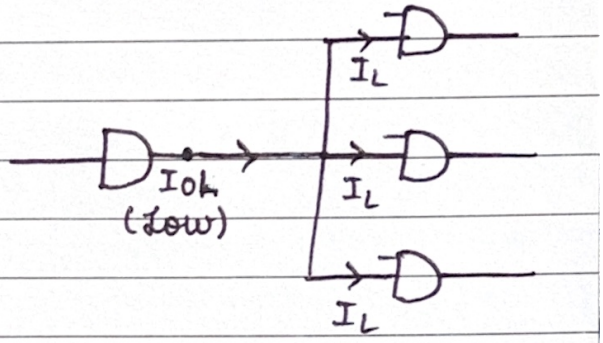
Fig of merit = Pd x tp → good IC always have low figure of merit

- Fan out should be as high as possible so that we can drive large no of gates by single gate. It defines current supply for given gate.



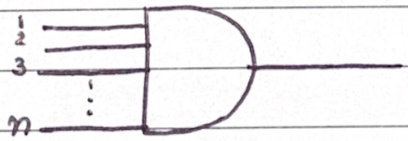
High state fan out

$$= \frac{I_{OH}\ (max)}{I_{IH}}$$



Low state fan out

$$= \frac{I_{OL}\ (min)}{I_{IL}}$$



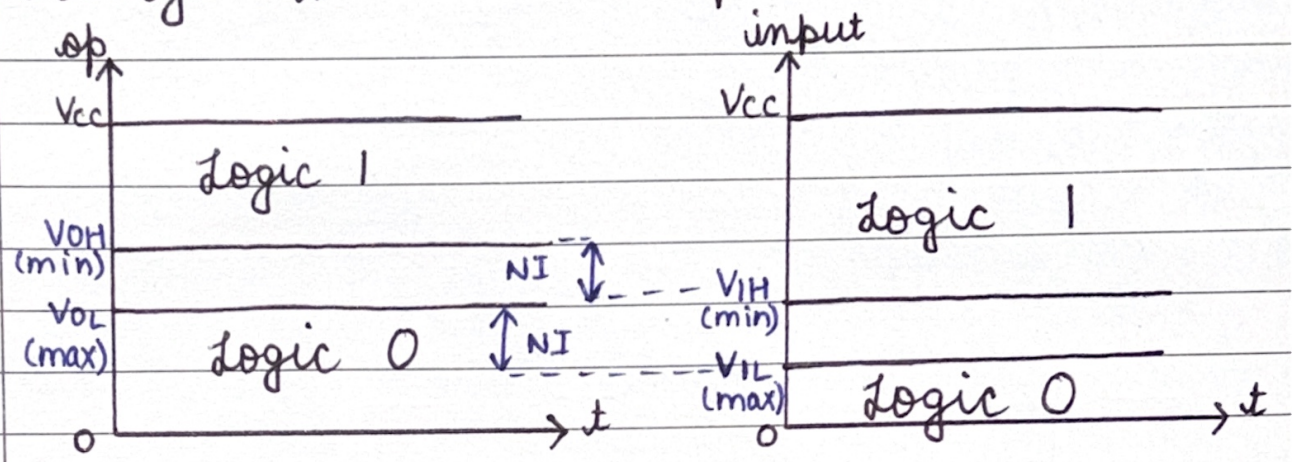
Fan In = n

7. Operating Temperature
   - For commercial applications (0-70°C)
   - For industrial applications (0-85°C)
   - For military applications (-55 - 125°C)

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→ Voltage and current parameters



- V_{OH} (min) is the minimum output voltage for logic 1
- V_{OL} (max) is the maximum output voltage for logic 0
- V_{IL} (max) is maximum input voltage for logic 0
- V_{IH} (min) is minimum input voltage for logic 1
- I_{OH} → high level output current
- I_{OL} → low level output current
- I_{IH} → high level input current
- I_{IL} → low level input current

V_{OH} (min) − V_{IH} (min) → Noise immunity for logic high

V_{IL} (max) − V_{OL} (max) → Noise immunity for logic low

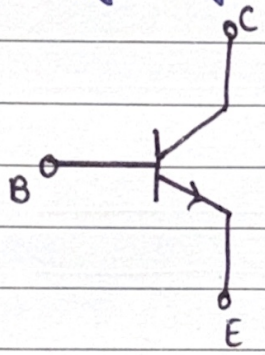
→ Resistor - Transistor Logic (RTL)

- This logic family includes resistors and transistors in it's IC. So it's called as RTL family
- Here we use transistor in saturated &

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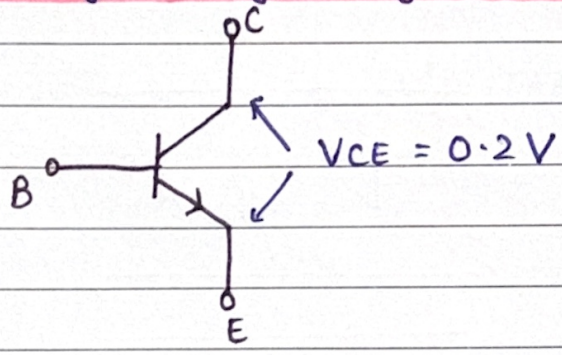
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cut off region, so speed of RTL family is low



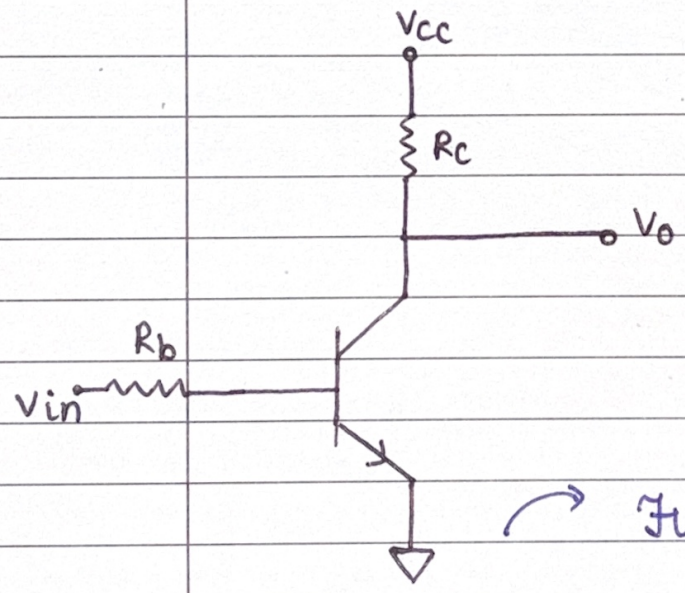
B = 0

cut off region
C to E terminal act like open circuit



B = Vcc

Saturation region
current flow from C to E



Input	output	Region
Low	High	cut off
High	Low	Saturation

Functions as NOT gate

→ Disadvantages of RTL

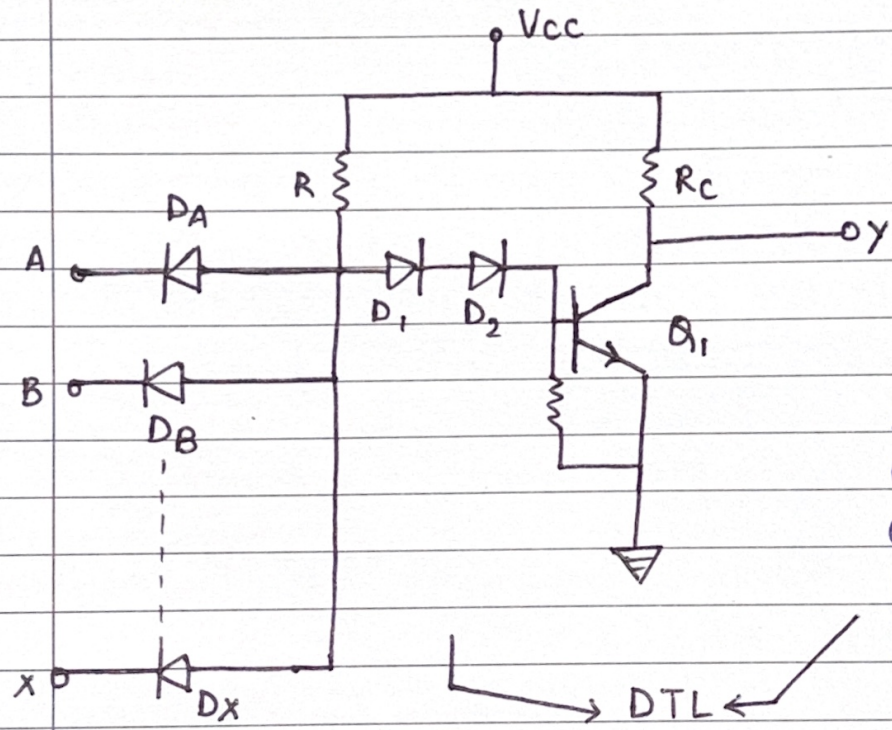
- Low noise margin
- Low fan out
- Low speed
- High power dissipation due to presence of resistors

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→ Diode Transistor Logic (DTL)

- RTL has low noise margin, low fan out, slow speed & higher power dissipation, so we don't use RTL in recent IC's
- DTL has improved noise margin & fan out compared to RTL



A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

DTL

circuit for NAND gate

→ Disadvantages of DTL

- Lower speed
- Power dissipation is high due to use of resistors

→ Transistor Transistor Logic (TTL)

1. Standard TTL
2. TTL totempole output
3. TTL open collector output
4. TTL tristate

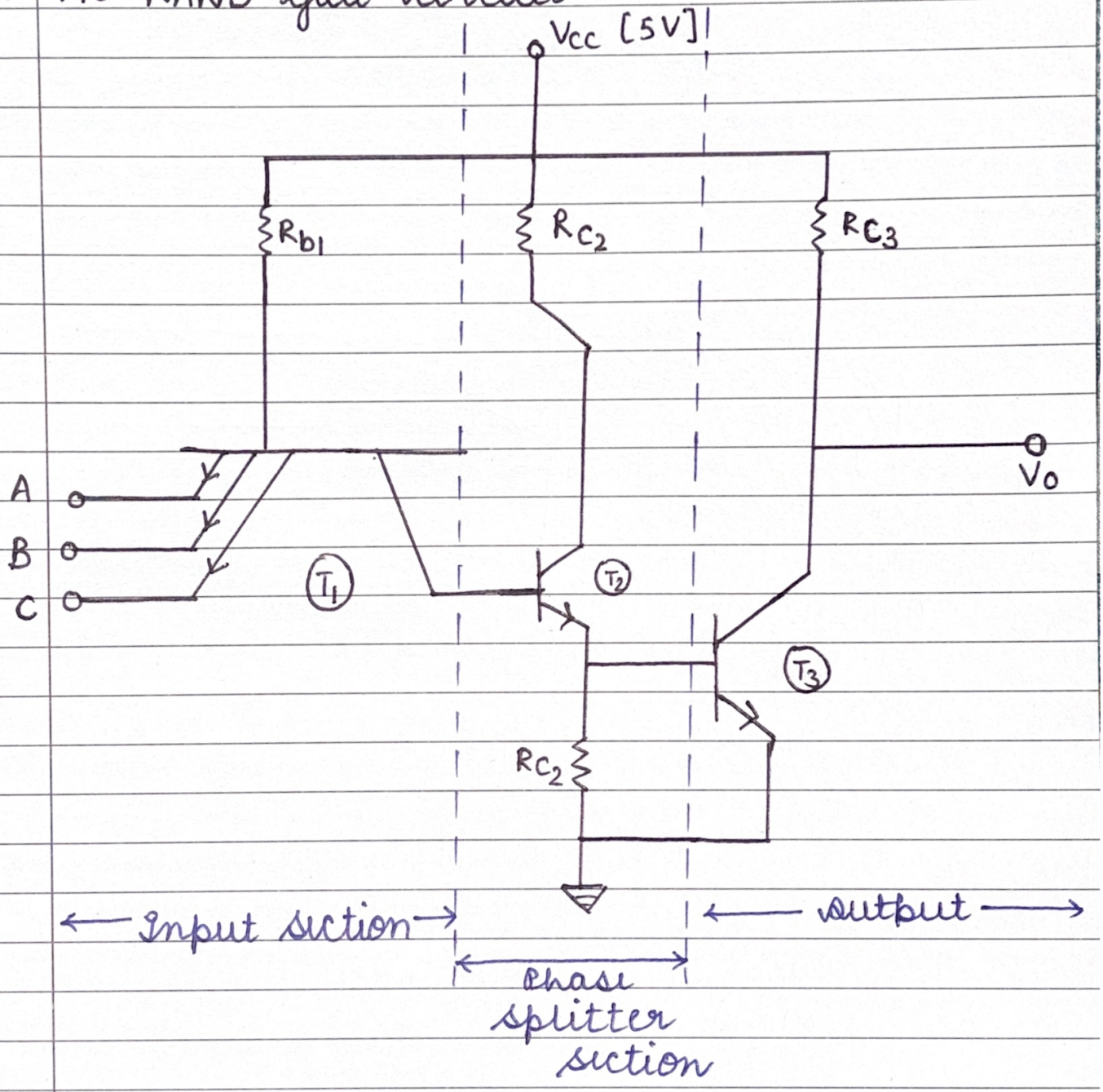
} classification of TTL

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- TTL offers way higher spud as compared to DTL and RTL

★
→ TTL NAND gate circuit



A	B	C	V_o
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1

A	B	C	V_o
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

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→ ☆ Working mechanism of TTL NAND gate

1. Case I → Atleast one output is low
   - Connecting any input to 0V forward biases the Base emitter junction of T1
   - Base current from Vcc flows out via T_1's emitter to ground
   - No base current reaches phase splitter so it remains off
   - Since phase splitter is off, no current flows into T_2's base so T_2 also remain off
   - Phase splitter's collector voltage rises towards +Vcc turning on T_3
   - Output node connects to +Vcc through T_4 & D_1
   - output Vo is high

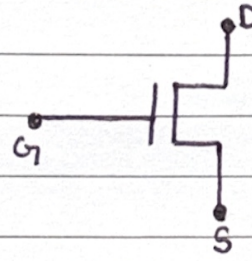
2. Case II → Both inputs are high
   - Both high inputs reverse bias both Base emitter junctions of T_1
   - Current from Vcc flows through T_1's base collector junction directly into base of phase splitter
   - Phase splitter is saturated & turns fully ON
   - Voltage drop across resistor R_{C_2} drives T_2's base turning T_2 fully on.
   - Phase splitter's collector voltage drops to ~0.9V which is too low to turn on T_3 thus T_3 turns off
   - output Vo is low

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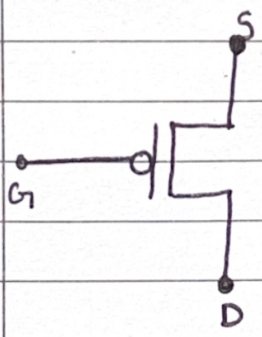
→ CMOS

CMOS stands for complimentary metal oxide semiconductor field effect transistor



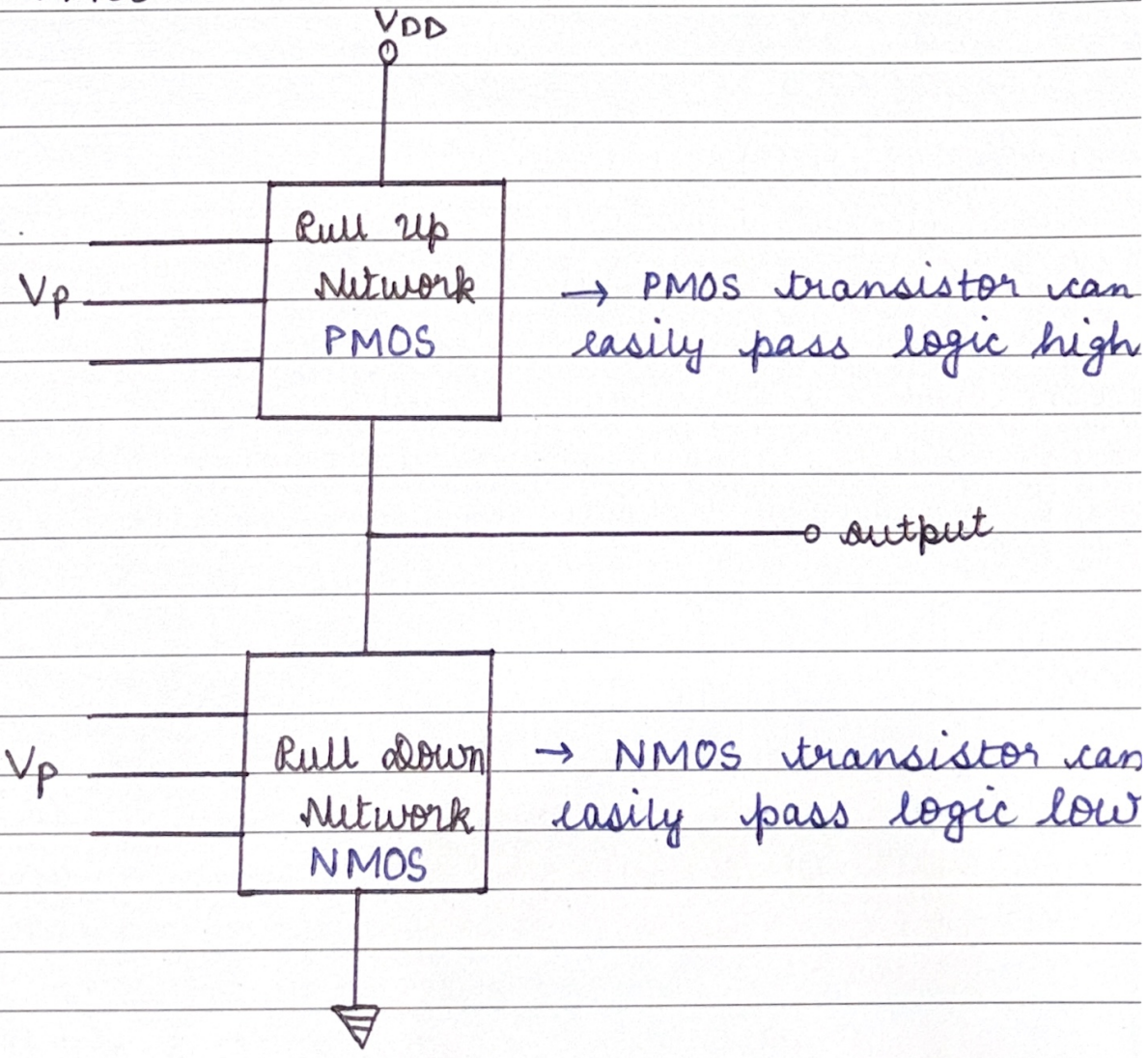
NMOS

- If G = 0, OFF, D to S acts as open circuit
- If G = 1, ON, D to S acts as short circuit



PMOS

- If G = 0, ON, S to D acts as short circuit
- If G = 1, OFF, S to D acts as open circuit



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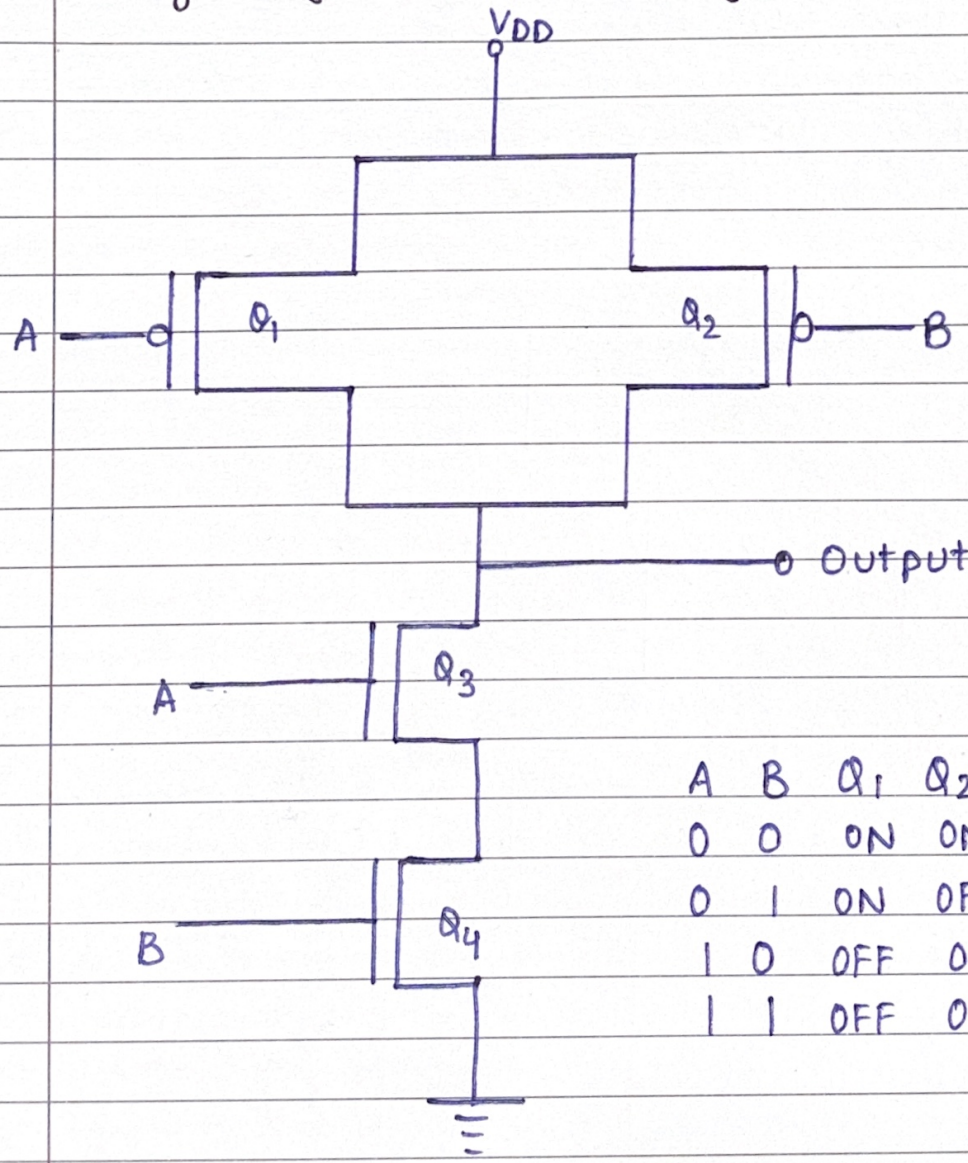
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→ Rules for logic circuits using CMOS

1. (·) operator [AND]
   - PMOS → Parallel
   - NMOS → Series

2. (+) operator [OR]
   - PMOS → Series
   - NMOS → Parallel

→ Design of CMOS NAND gate



A	B	Q_1	Q_2	Q_3	Q_4	Y
0	0	ON	ON	OFF	OFF	1
0	1	ON	OFF	OFF	ON	1
1	0	OFF	ON	ON	OFF	1
1	1	OFF	OFF	ON	ON	0

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→ Interfacing of different logic families

- An interface is a circuit that connects two different logic families
- Interfacing means connecting outputs of one circuit / system to inputs of system with different electrical characterstics

→ Why interfacing is required ?

- There are a number of logic families each having it's own strong points.
- In designing more complex digital systems the designers utilize different logic families for different parts of system in order to take advantage of strong points of each family

1. TTL to ECL interface

TTL → low speed but widely used.

ECL → fastest, handles high data routes

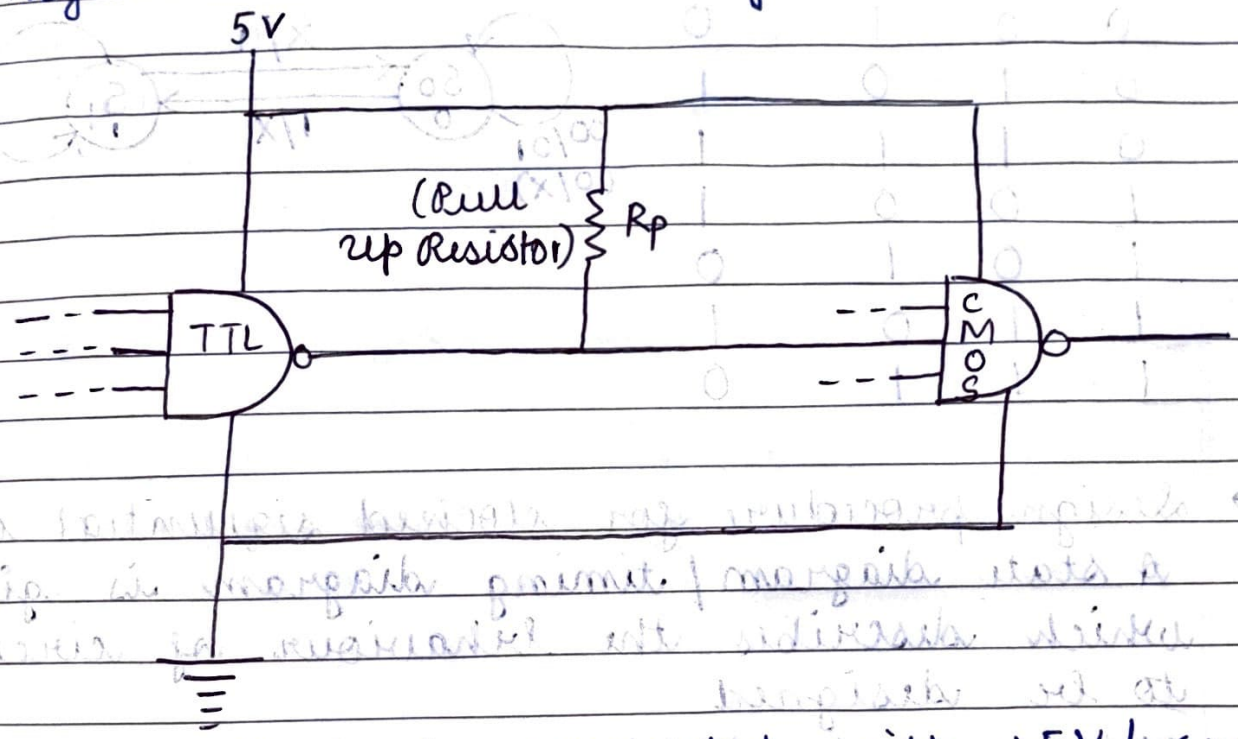
TTL	ECL
0V → low	-1.6V → low
5V → high	-0.8V → high

↳ We use TTL-ECL interface which is a level convertor circuit & changes TTL's 0/5V to ECL's -1.6/-0.8V

- Used when input data rate is slow but output is fast

2. TTL to CMOS interfacing

TTL	CMOS
logic 1 → 2·4 – 5 V	logic 1 = 5V / 10V
logic 0 → 0 – 0·8 V	logic 0 = 0V

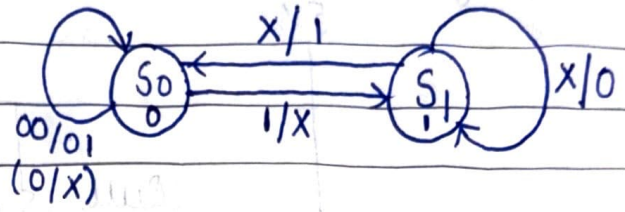


- TTL output is connected with +5V / source with a pull up resisor which causes the TTL output to rise to approximately +5V in high state thereby proving adequate input, turns on CMOS

→ State diagram

- For JK flip flop

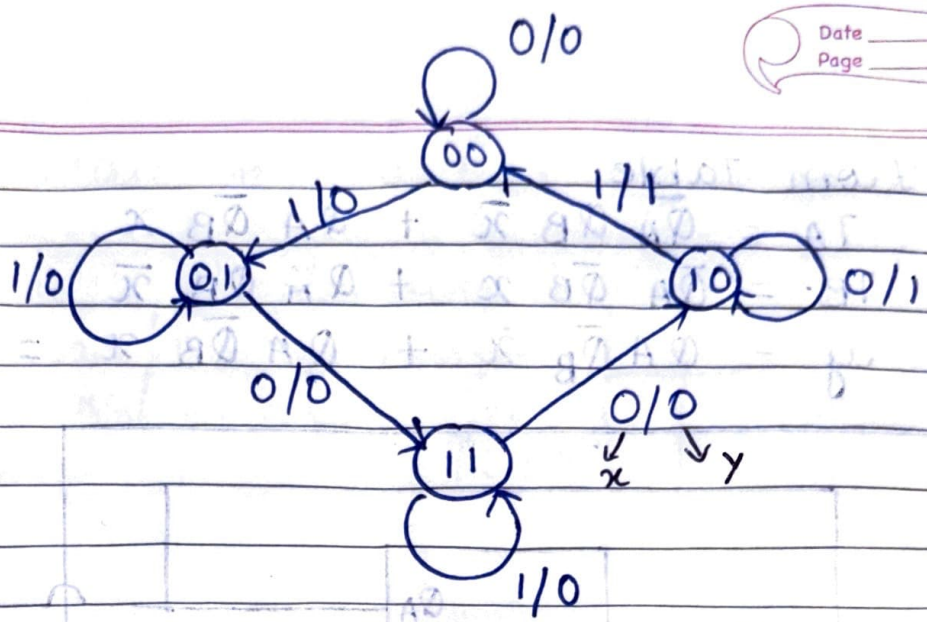
Q_n	J	K	Q_n^+
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0



→ Design procedure for clocked sequential circuits

1. A state diagram / timing diagram is given which describes the behaviour of circuit to be designed
2. Obtain state table
3. No of states can be reduced by state reduction method
4. Do state assignment (if required)
5. Determine the no. of flip flops required and assign letter symbols
6. Decide the type of flip flops to be used
7. Derive circuit excitation table from the state table
8. Obtain expression for circuit output and flip flop input
9. Implement the circuit

For eg :-



Present Q_A Q_B	Next state $x=0$ $Q_A^+ Q_B^+$	Next state $x=1$ $Q_A^+ Q_B^+$	output Y $x=0$	output Y $x=1$
0 0	0 0	0 1	0	0
0 1	1 1	0 1	0	0
1 0	1 0	0 0	1	1
1 1	1 0	1 1	0	0

Check next state & outputs, if same keep one & remove others → state reduction

Here, no repetition so reduction X

4 states ⇒ 2 flip flops required → A & B

Circuit excitation Table T_A T_B

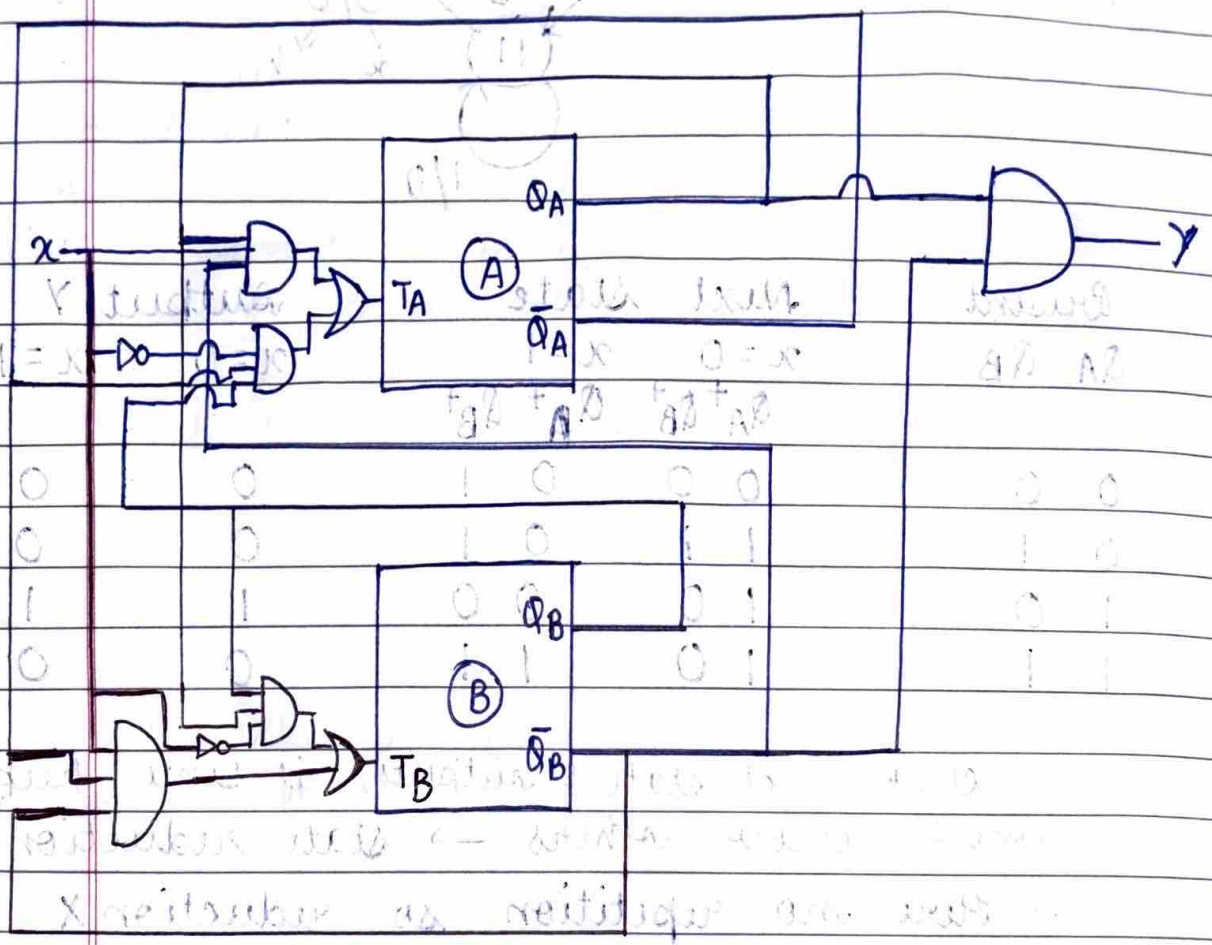
Q_A	Q_B	x	Q_A^+	Q_B^+	FF inputs		Y
0	0	0	0	0	0	0	0
0	0	1	0	1	0	1	0
0	1	0	1	1	1	0	0
0	1	1	0	1	0	0	0
1	0	0	1	0	0	0	1
1	0	1	0	0	1	0	1
1	1	0	1	0	0	1	0
1	1	1	1	1	0	0	0

From Table

$$T_A = \bar{Q}_A Q_B \bar{x} + Q_A \bar{Q}_B x$$

$$T_B = \bar{Q}_A \bar{Q}_B x + Q_A Q_B \bar{x}$$

$$y = Q_A \bar{Q}_B \bar{x} + Q_A \bar{Q}_B x = Q_A \bar{Q}_B$$



→ Mealy & Moore state machines

There are two models developed for representing synchronous sequential circuits

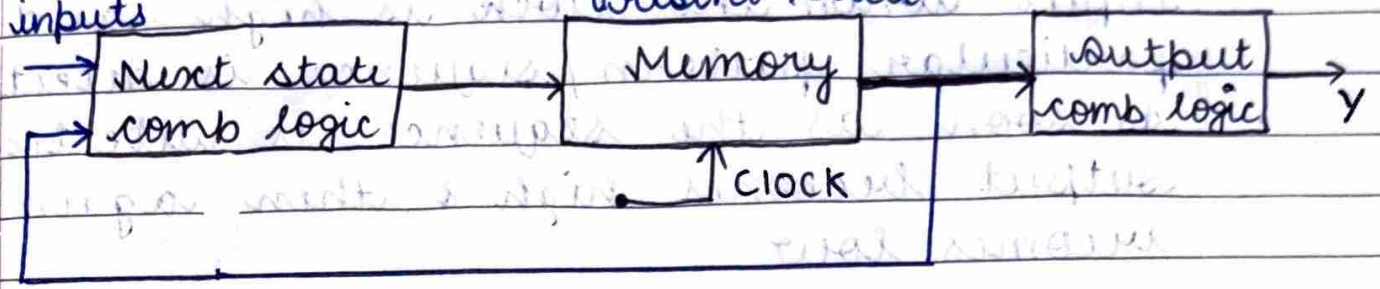
1. MOORE state machine
2. MEALEY state machine

→ only difference is the way in which output is generated

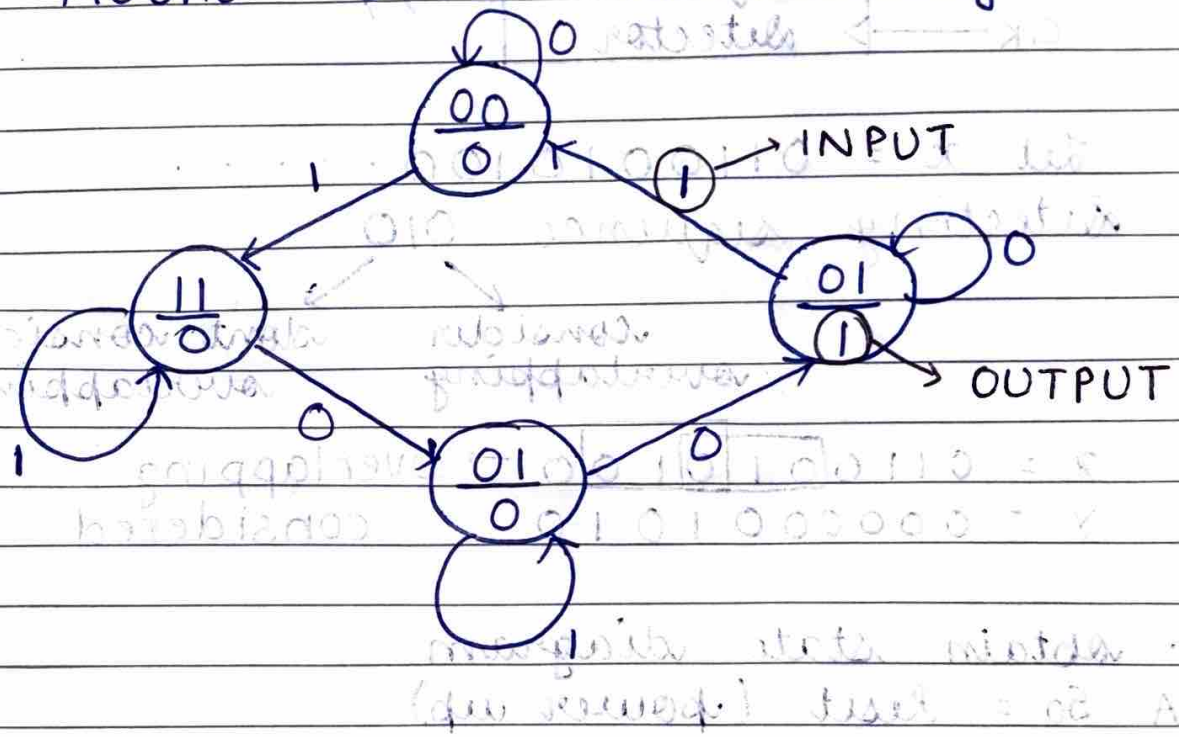
Moore → output is the function of the present state only

Mealey → output is the function of the present state as well as the inputs

→ Moore state machine

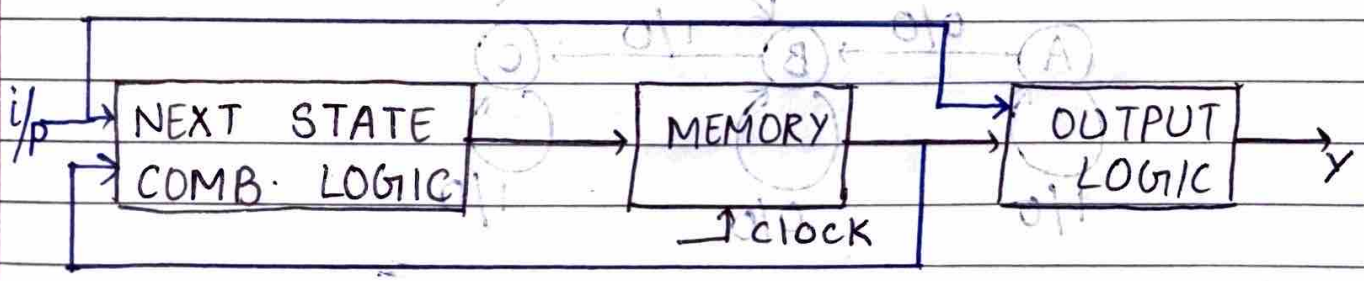


MOORE STATE DIAGRAM eg :-



→ MEALEY STATE MACHINES

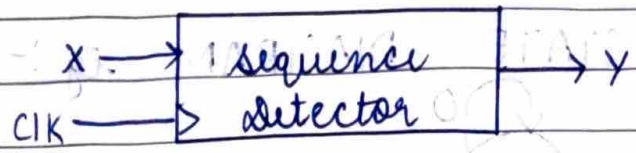
The output is function of present state as well as the input.



No of states required in mealey machines are less as compared to moore machines

→ Sequence Detector

- The stream of bits has been fed as the input when the clock is high and a particular pattern / sequence is detected.
- As soon as the sequence is detected, the output becomes high & then again it becomes low



Let x = 0110010100 . . . . .

Detecting sequence 010

consider overlapping — dont consider overlapping

x = 0 1 1 0 0 1 0 1 0 0 → overlapping considered

y = 0 0 0 0 0 0 1 0 1 0

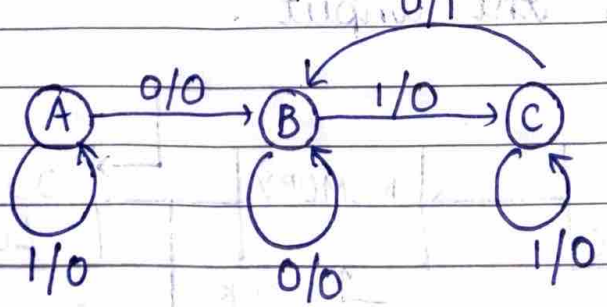
1. Obtain state diagram

A S_0 = Reset (power up)

B S_1 = 0

C S_2 = 01

Ø/0
Ø/1



Input	PS	NS	output
0	A	B	0
1	A	A	0
0	B	B	0
1	B	C	0
0	C	B	1
1	C	C	0

→ How to do state assignment

1. Same next state

Agr 2 ya usse jyada states kisi ek same input pr same N.S de to vo adjacent hongi (1 bit difference)

eg. $A \xrightarrow{0} B$ & $B \xrightarrow{0} B$

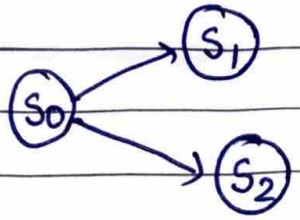
⇒ A & B must be adjacent

$B \xrightarrow{1} C$ & $C \xrightarrow{1} C$

⇒ B & C must be adjacent

2. Same successor state

If



then S_1 & S_2 must have adjacent codes

3. Same outputs

Jo state same input pe same output deti hai unko bi adjacent dene ka try kro

So now

A = 00

B = 10

C = 11

A adj B

& B adj C

	Input	PS		NS		Output		
	X	Q_1	Q_0	Q_1^+	Q_0^+	Y	T_1	T_0
0	0	0	0	0	1	0	0	1
1	0	0	1	0	1	0	0	0
3	0	1	1	0	1	1	1	0
4	1	0	0	0	0	0	0	0
5	1	0	1	1	1	0	1	0
7	1	1	1	1	1	0	0	0

K-map for Y

	$\bar{Q}_1\bar{Q}_0$	$\bar{Q}_1 Q_0$	$Q_1 Q_0$	$Q_1\bar{Q}_0$
$\bar{X}$	0	0	1	X
X	0	0	0	X

$Y = \bar{X} Q_1$

Kmap for T_1

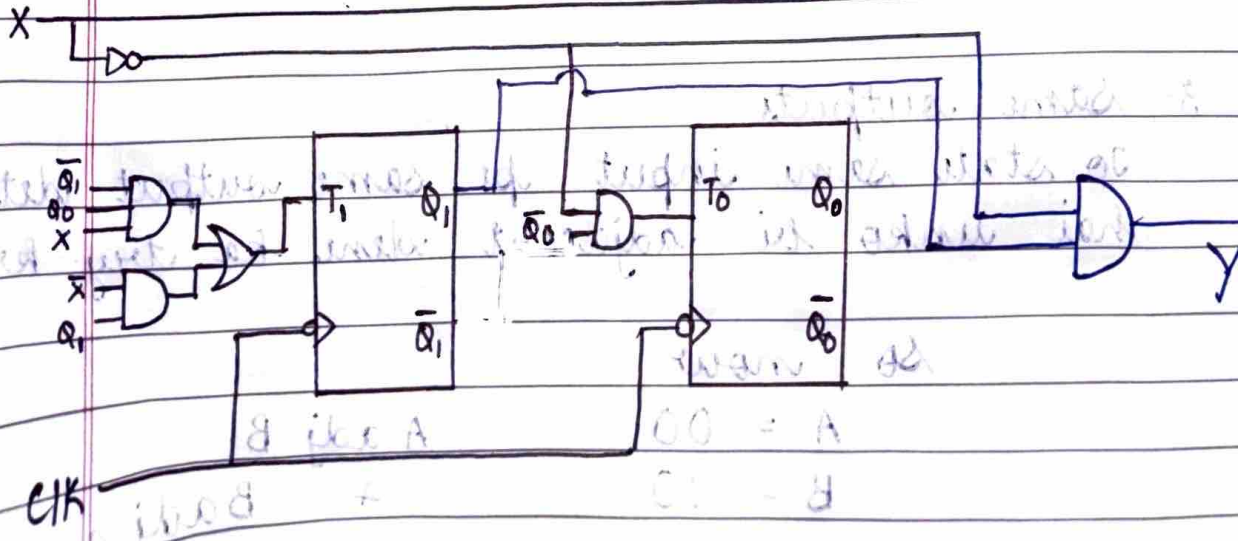
	0	0	1	X
	0	1	0	X

$T_1 = \bar{Q}_1 Q_0 X + \bar{X} Q_1$

K-map for T_0

1	0	0	X
0	0	0	X

$T_0 = \bar{X}\bar{Q}_0$



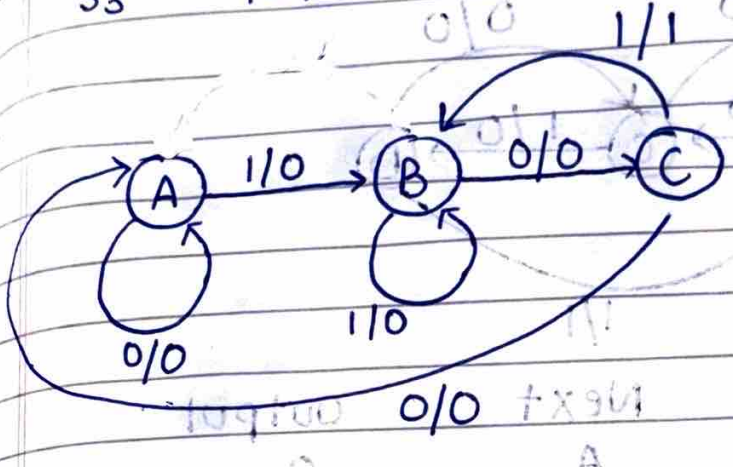
101 detector

S_0 → Reset (A)

S_1 → 1 (B)

S_2 → 10 (C)

S_3 → 101 (Z=1)



Present	X	Next	Z
A	0	A	0
A	1	B	0
B	0	C	0
B	1	B	0
C	0	A	0
C	1	B	1

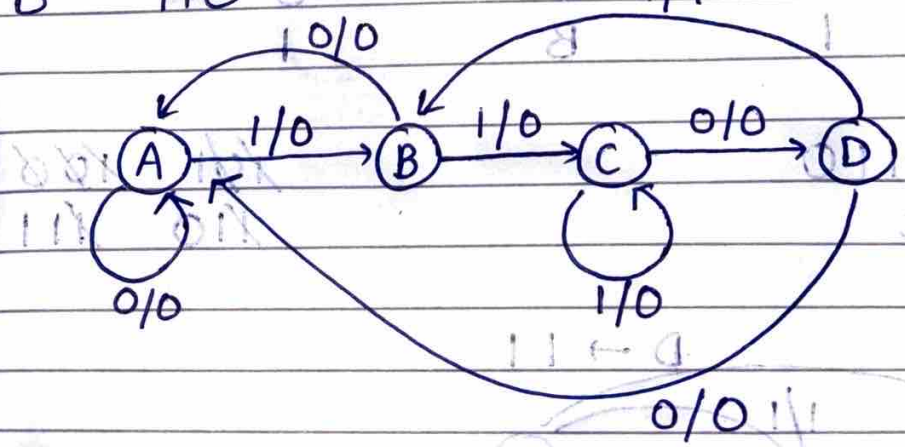
1101 detector

A → RESET

B → 1

C → 11

D → 110



current	I/P	NEXT	O/P
A	0	A	0
A	1	B	0
B	0	A	0
B	1	C	0
C	0	D	0
C	1	C	0
D	0	A	0
D	1	B	1

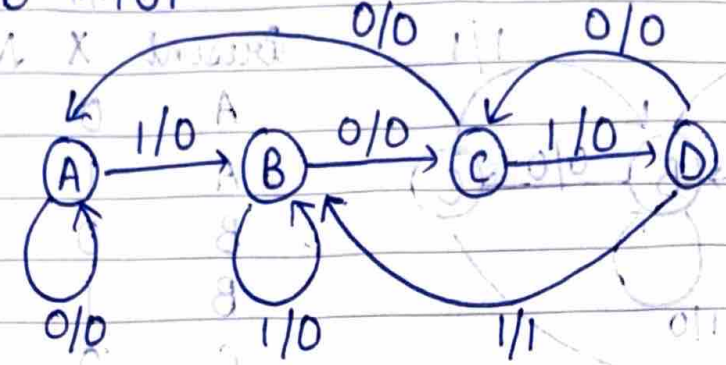
→ 1011 detector

A → Reset

B → 1

C → 10

D → 101



Present	input	Next	output
A	0	A	0
A	1	B	0
B	0	C	0
B	1	B	0
C	0	A	0
C	1	D	0
D	0	C	0
D	1	B	1

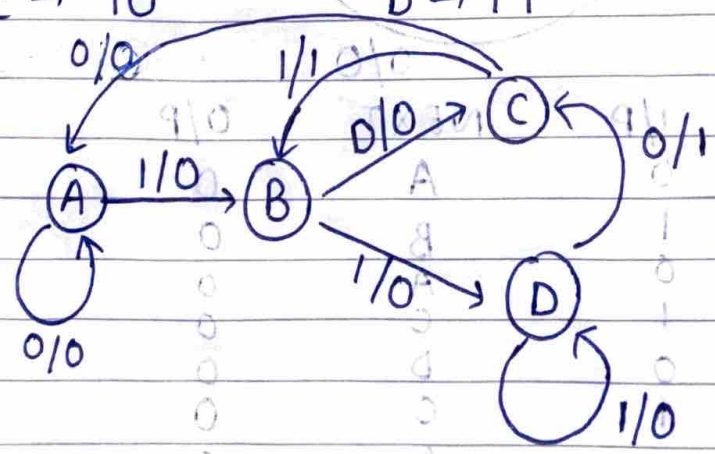
→ 101 or 110

A → Reset

B → 1

C → 10

D → 11



A	0	A	0
A	1	B	0
B	0	C	0
B	1	D	0
C	0	A	0
C	1	B	1
D	0	C	1
D	1	D	0

1001 or 0110

A → Reset

B → 1

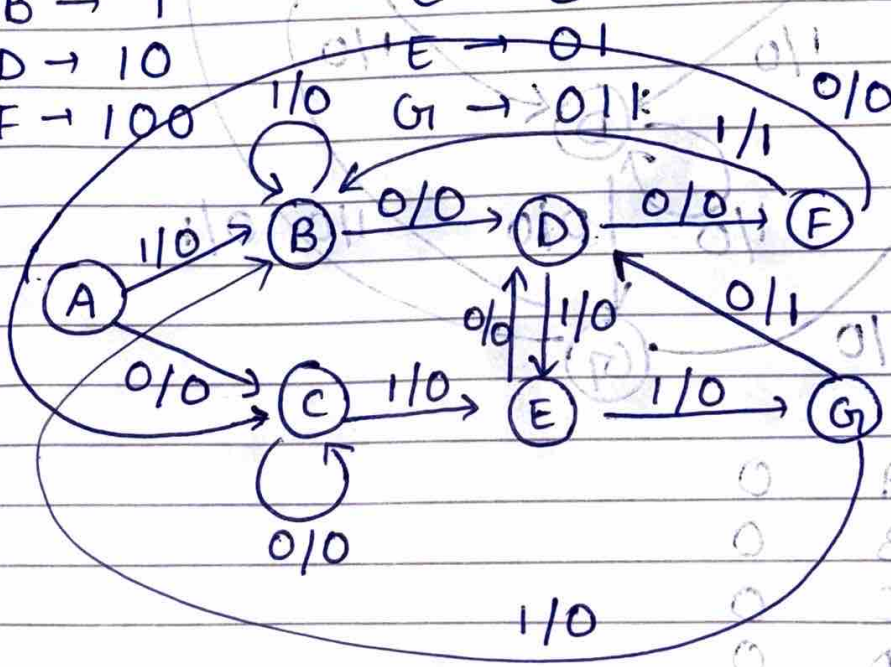
D → 10

F → 100

C → 0

E → 01

G → 011



A	0	C	0
A	1	B	0
B	0	D	0
B	1	B	0
C	0	C	0
C	1	E	0
D	0	F	0
D	1	E	0

E	0	D	0
E	1	G	0
F	0	C	0
F	1	B	1
G	0	A	1
G	1	B	0

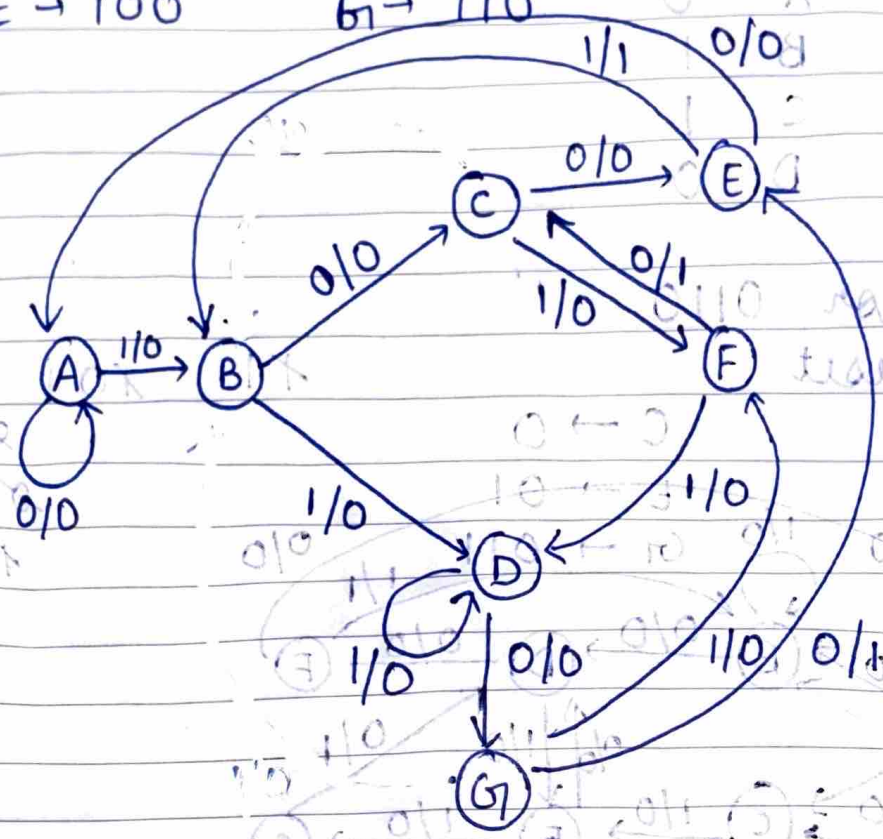
→ 1100 or 1001 or 1010

A → RESET

B → 1

C → 10 D → 11

F → 101 E → 100 G → 110



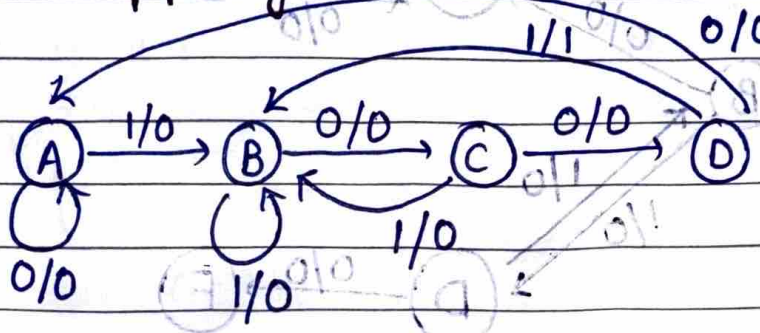
A	0	A	0
A	1	B	0
B	0	C	0
B	1	D	0
C	0	E	0
C	1	F	0
D	0	G	0
D	1	D	0
E	0	A	0
E	1	B	1
F	0	C	1
F	1	D	0
G	0	E	1
G	1	F	0

→ Difference between overlapping & non-overlap.

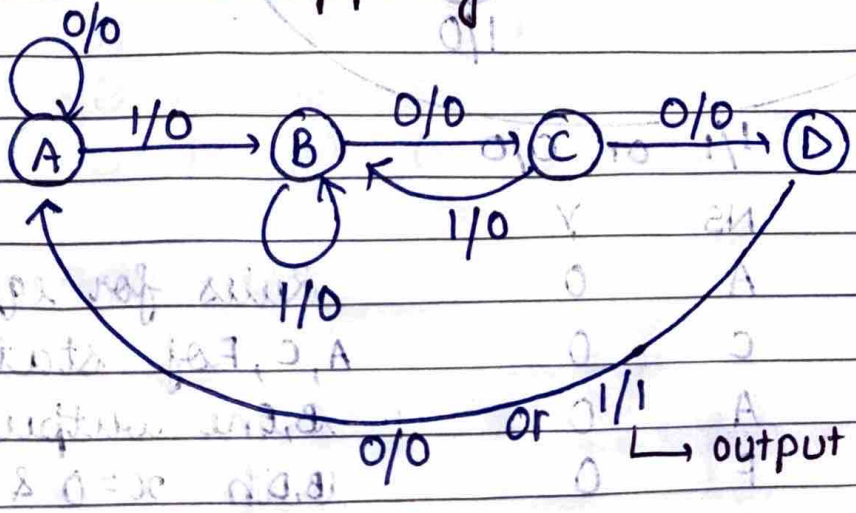
Non overlapping sequence me jese hi output 1 hoga, reset hojayga jabki overlap me kisi pichli state pe jaata hai

eg → 1001 ← A = Reset B = 1 C = 10
D = 100

overlapping :-



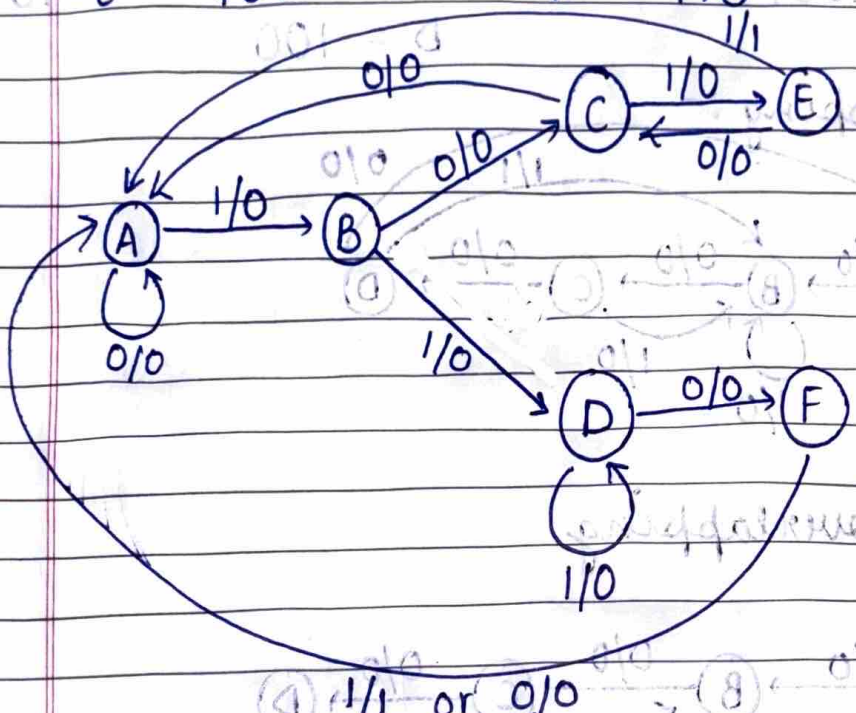
Non - overlapping



PYQ

Q Design a sequence detector to detect the non overlapping sequences 1011 & 1101. It generates output 1 when sequence is detected

A → Reset
B → 1
C → 10
D → 11
E → 1011
F → 1101



1/1 or 0/0

X	PS	NS	Y
0	A	A	0
0	B	C	0
0	C	A	0
0	D	F	0
0	E	C	0
0	F	A	0
1	A	B	0
1	B	D	0
1	C	E	0
1	D	D	0
1	E	A	1
1	F	A	1

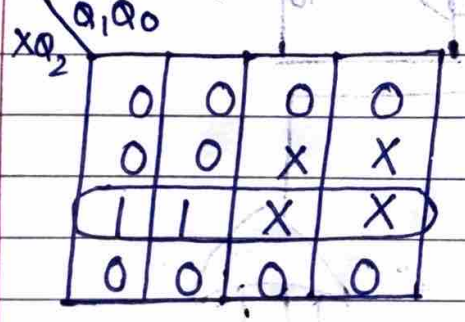
Rules for equalency of states :-
1. same outputs for both $x=0$ & $x=1$ inputs
2. same next states

→ No equivalent states

A = 000
B = 001
C = 010
D = 011
E = 100
F = 101

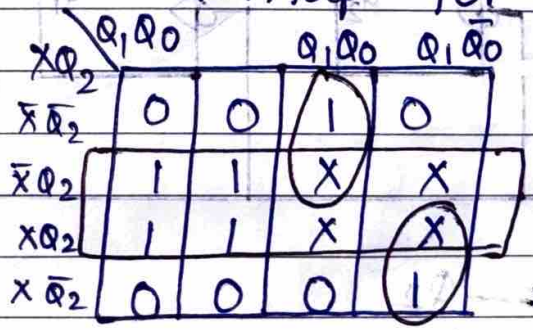
X	Q_2	Q_1	Q_0		Q_2^+	Q_1^+	Q_0^+	Y	T_2	T_1	T_0
0	0	0	0	(0)	0	0	0	0	0	0	0
0	0	0	1	(1)	0	1	0	0	0	1	1
0	0	1	0	(2)	0	0	0	0	0	1	0
0	0	1	1	(3)	1	0	1	0	1	1	0
0	1	0	0	(4)	0	1	0	0	1	1	0
0	1	0	1	(5)	0	0	0	0	0	0	1
1	0	0	0	(8)	0	0	1	0	0	0	1
1	0	0	1	(9)	0	1	1	0	0	1	0
1	0	1	0	(10)	1	0	0	0	1	1	0
1	0	1	1	(11)	0	1	1	0	0	0	0
1	1	0	0	(12)	0	0	0	1	1	0	0
1	1	0	1	(13)	0	0	0	1	1	0	1

K-map for Y



$\Rightarrow Y = Q_2 X$

K map for T_2



$T_2 = Q_2 + Q_1\bar{Q}_0 X + Q_1 Q_0 \bar{X}$
$= Q_2 + Q_1 (Q_0 \oplus X)$

K-map for T_1

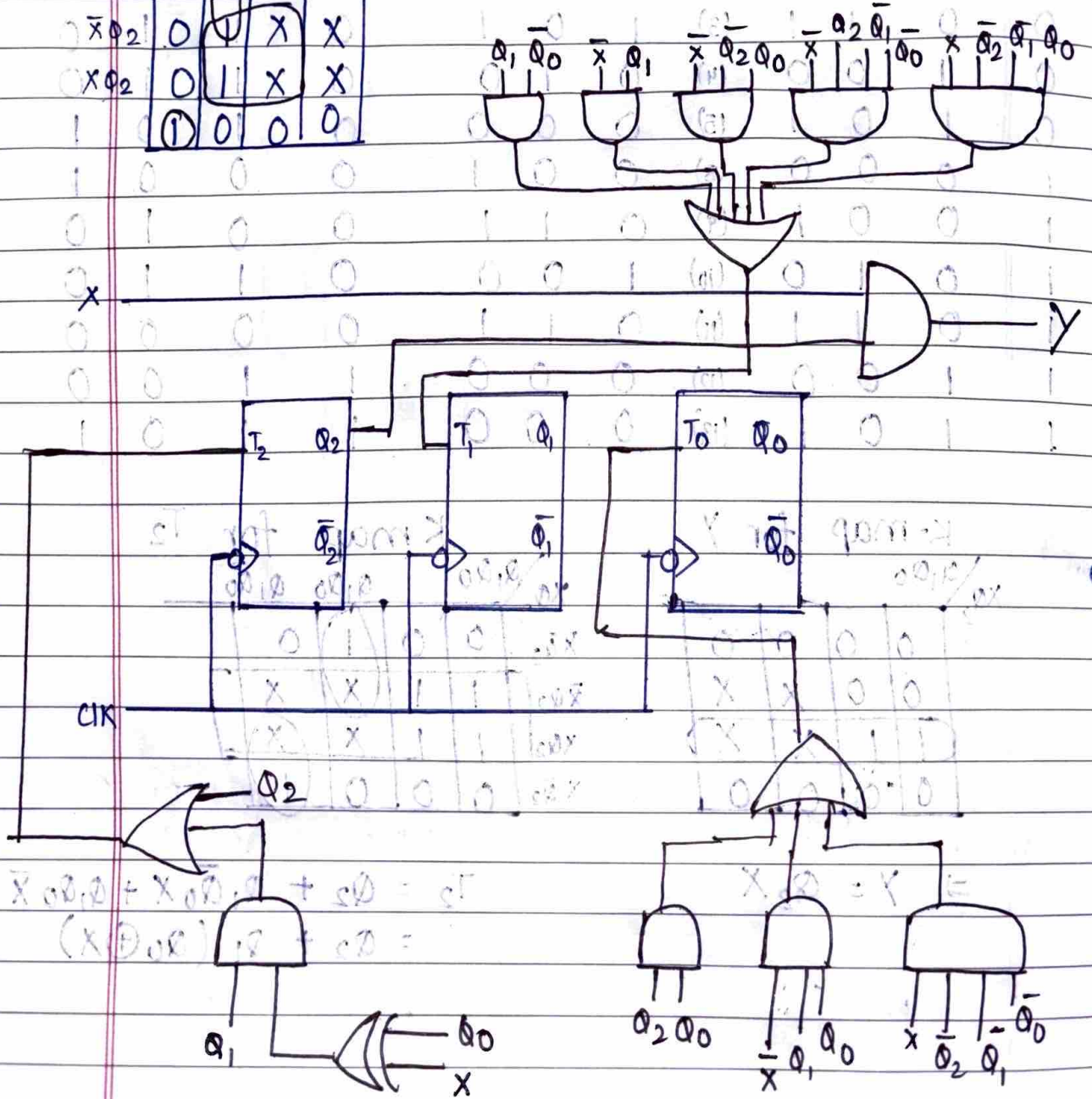
xQ_2 \ Q_1Q_0	00	01	11	10
0	1	1	1	1
1	0	X	X	0
0	0	X	X	0
0	1	0	1	0

$$T_1 = Q_1\bar{Q}_0 + \bar{x}Q_1 + \bar{x}\bar{Q}_2Q_0 + \bar{x}Q_2\bar{Q}_1\bar{Q}_0 + xQ_2\bar{Q}_1Q_0$$

Kmap for T_0

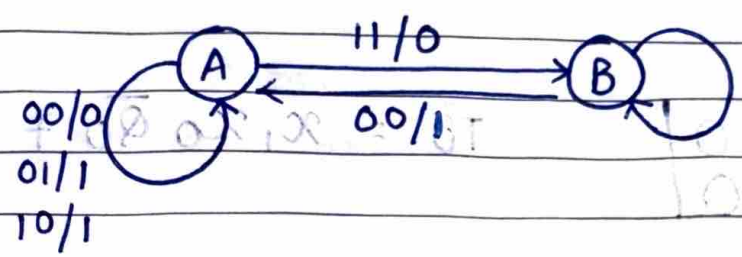
	$\bar{Q}_1\bar{Q}_0$	$\bar{Q}_1Q_0$	Q_1Q_0	$Q_1\bar{Q}_0$
$\bar{x}\bar{Q}_2$	0	1	0	0
$\bar{x}Q_2$	0	1	X	X
xQ_2	0	1	X	X
$x\bar{Q}_2$	1	0	0	0

$$T_0 = Q_2Q_0 + \bar{x}\bar{Q}_1Q_0 + x\bar{Q}_2\bar{Q}_1\bar{Q}_0$$



→ Serial Binary Adder

A → carry 0 was generated from prev add^n
B → carry 1 was generated from prev add^n

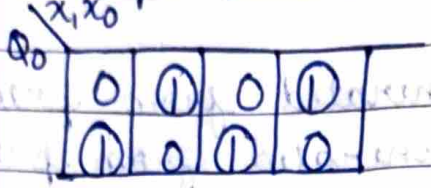


x_1	x_0	PS	NS	Y
0	0	A	A	0
0	1	A	A	1
1	0	A	A	1
1	1	A	B	0
0	0	B	A	1
0	1	B	B	0
1	0	B	B	0
1	1	B	B	1

Let A = 0 & B = 1

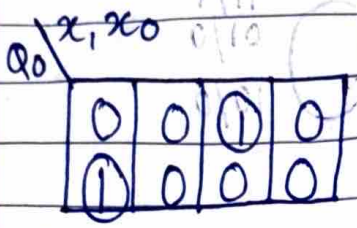
Q_0 PS	x_1	x_0	Q_0^+ NS	Y	T_0
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	1	0
0	1	1	1	0	1
1	0	0	0	1	1
1	0	1	1	0	0
1	1	0	1	0	0
1	1	1	0	1	1

Kmap for Y

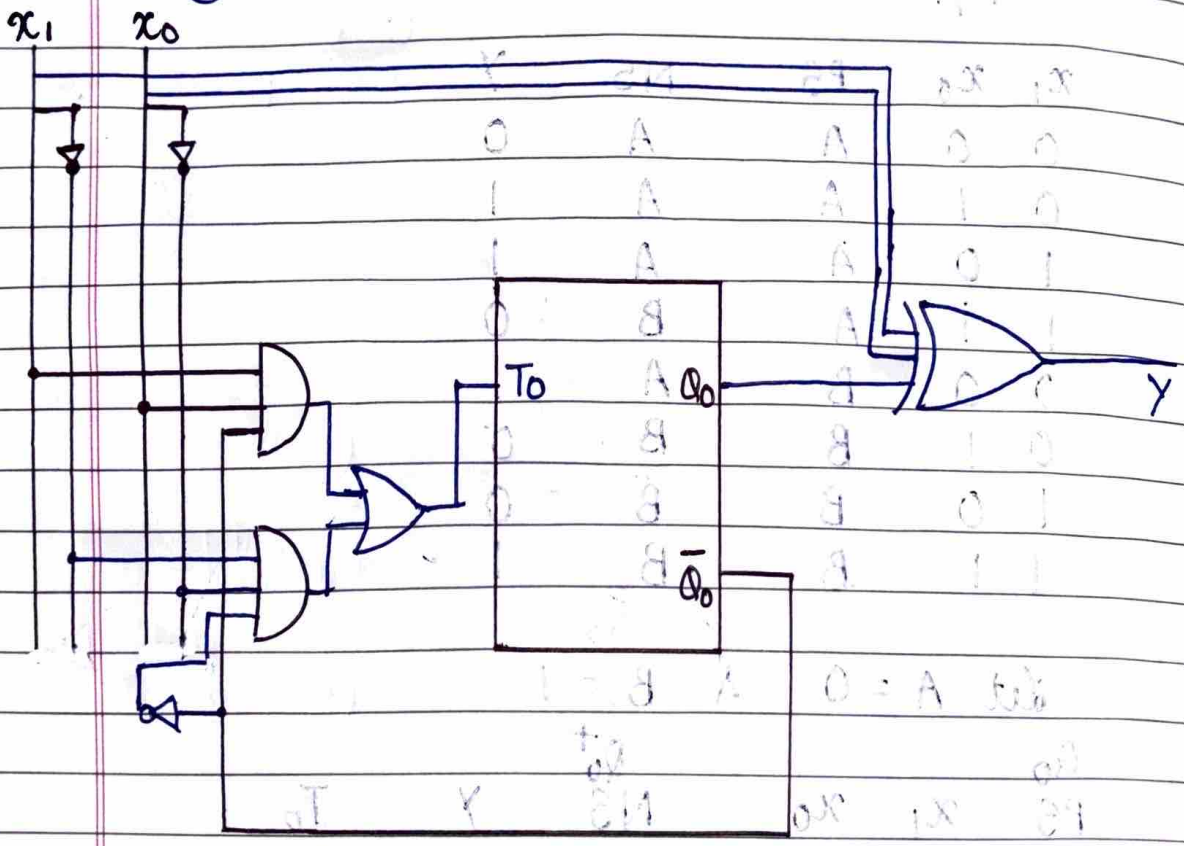


$$Y = Q_0 \oplus x_1 \oplus x_0$$

Kmap for T_0

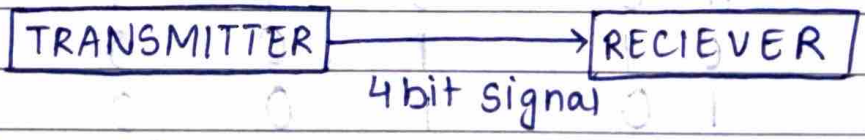


$$T_0 = x_1 x_0 \bar{Q}_0 + Q_0 \bar{x}_1 \bar{x}_0$$



→ Parity

It's a concept to detect a single bit error



Along with the 4 bit stream of signal

we also send an extra bit that tells us about the total number of 1's.

Parity → even
Parity → odd

0100 → Parity Bit
(original signal)

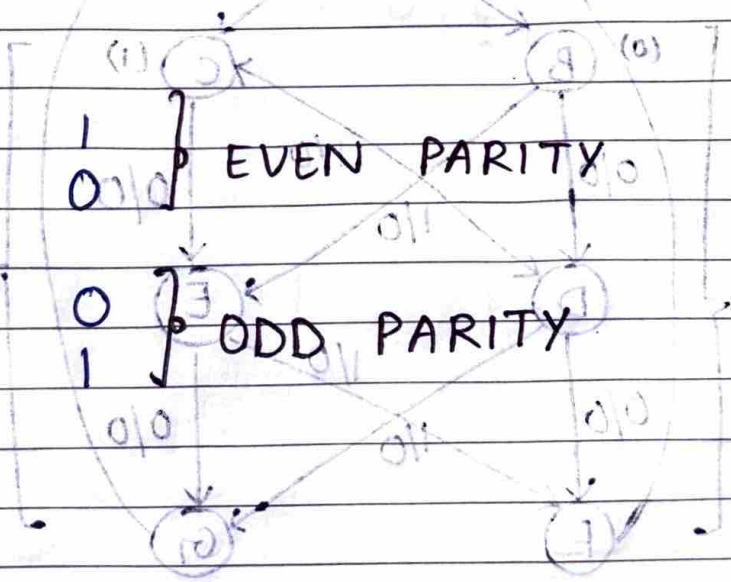
If signal me total 1's odd h (eg → 0100) aur even parity chahiye to parity bit ki value 1 hogi taki total 1's even rahe

Isi trh odd parity me bit ki value 0 rkhege taki total 1's odd rahe.

eg : 0100 = 7

0 0 0 1 0 1 0 0 → 1 } EVEN PARITY
1 1 0 0 → 0 }

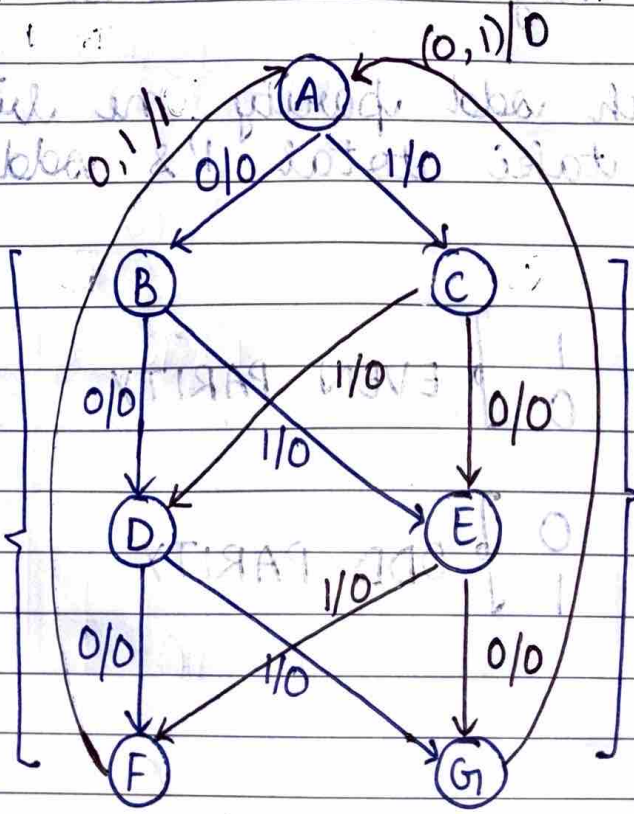
0 1 0 0 → 0 } ODD PARITY
1 1 0 0 → 1 }



PYQ

→ Add Parity Bit generator by J & K, F.F.
3 bit

B_2	B_1	B_0	P
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0



B = 0 when C = 1
D = 00 or 11
E = 01 or 10

B, D, F → Even no of 1's

C, E, G → odd no. of 1's

B & C use krke LSB generate kri
D & E se second bit
F & G se MSB

PS	NS x=0	NS x=1	Y x=0	Y x=1
A	B	C	0	0
B	D	E	0	0
C	E	D	0	0
D	F	G	0	0
E	G	F	0	0
F	A	A	1	1
G	A	A	0	0

Q	Q^+	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

B & C → adj

D & E → adj

F & G → adj

A = 000

B = 010

C = 011

D = 110

E = 111

F = 100

G = 101

	P.S. X	B_2	B_1	B_0	N.S B_2^+	B_1^+	B_0^+	Y	J_2	K_2	J_1	K_1	J_0	K_0
A	0	0	0	0	0	1	0	0	0	X	1	X	0	X
	0	0	0	1	X	X	X	X	X	X	X	X	X	X
B	0	0	1	0	1	1	0	0	1	X	X	0	0	X
C	0	0	1	1	1	1	1	0	1	X	X	0	X	0
F	0	1	0	0	0	0	0	1	X	1	0	X	0	X
G	0	1	0	1	0	0	0	0	X	1	0	X	X	1
D	0	1	1	0	1	0	0	0	X	0	X	1	0	X
E	0	1	1	1	1	0	1	0	X	0	X	1	X	0
A	1	0	0	0	0	1	1	0	0	X	1	X	1	X
	1	0	0	1	X	X	X	X	X	X	X	X	X	X
B	1	0	1	0	1	1	1	0	1	X	X	0	1	X
C	1	0	1	1	1	1	0	0	1	X	X	0	X	1
F	1	1	0	0	0	0	0	1	X	1	0	X	0	X
G	1	1	0	1	0	0	0	0	X	1	0	X	X	1
D	1	1	1	0	1	0	1	0	X	0	X	1	1	X
E	1	1	1	1	0	0	0	0	X	0	X	1	X	1

Kmap for Y

$x B_2$ \ B_1B_0	00	01	11	10
0	X	0	0	0
1	0	0	0	0
1	0	0	0	0
0	X	0	0	0

$Y = \bar{B}_1 \bar{B}_0 B_2$

Kmap for J_2

0	X	1	1
X	X	X	X
X	X	X	X
0	X	1	1

$J_2 = B_1$

Kmap for K_2

X	X	X	X
1	1	0	0
1	1	0	0
X	X	X	X

$K_2 = \bar{B}_1$

Kmap for J_1

$\bar{x} B_2$	1	X	X	X
	0	0	X	X
	0	0	X	X
$x \bar{B}_2$	1	X	X	X

$J_1 = \bar{B}_2$

Kmap for K_1

	X	X	0	0
$\bar{x} B_2$	X	X	1	1
$x B_2$	X	X	1	1
	X	X	0	0

$K_1 = B_2$

Kmap for J_0

0	X	X	0
0	X	X	0
0	X	X	1
1	X	X	1

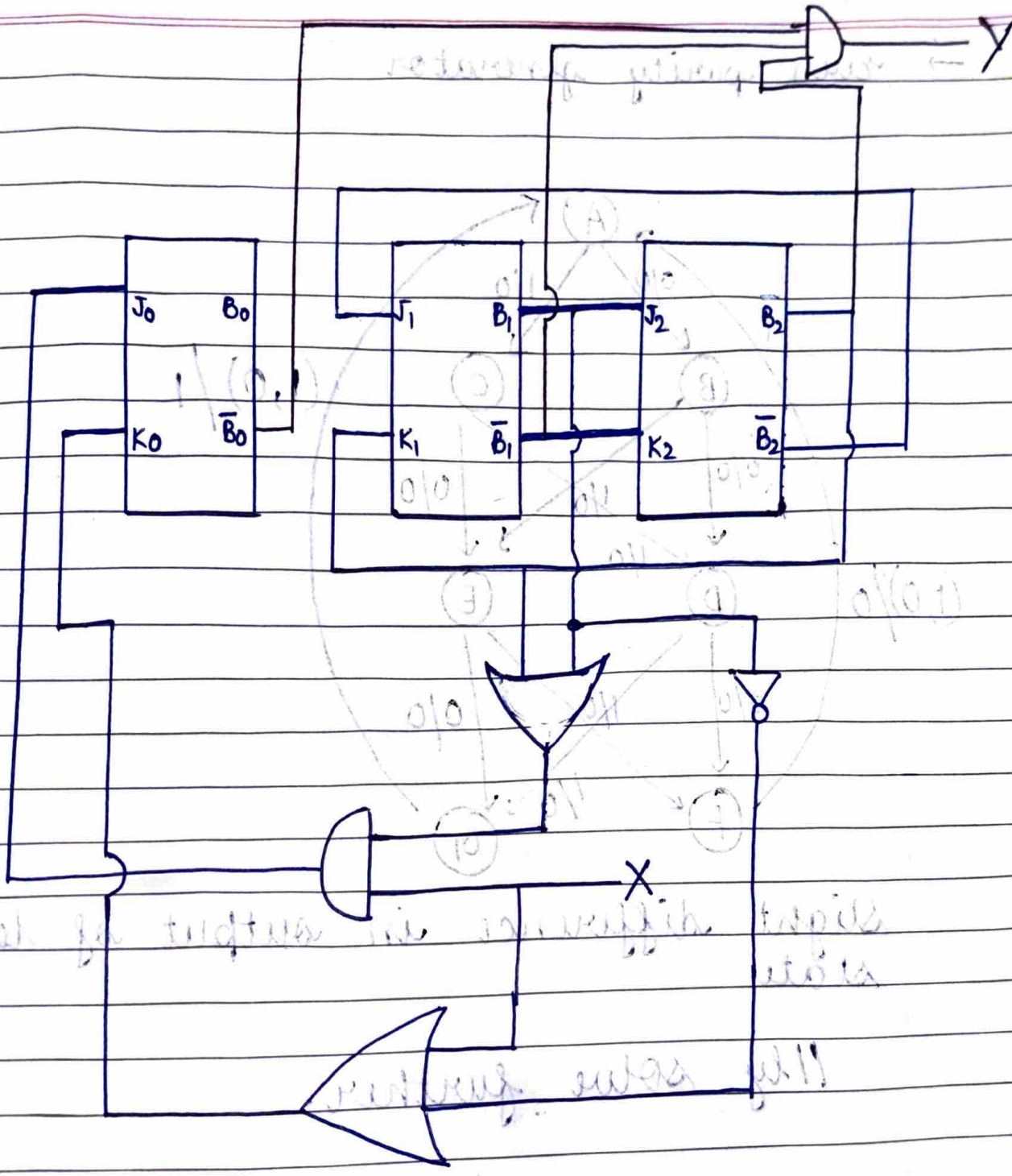
$J_0 = x \bar{B}_2 + \bar{x} B_1$

Kmap for K_0

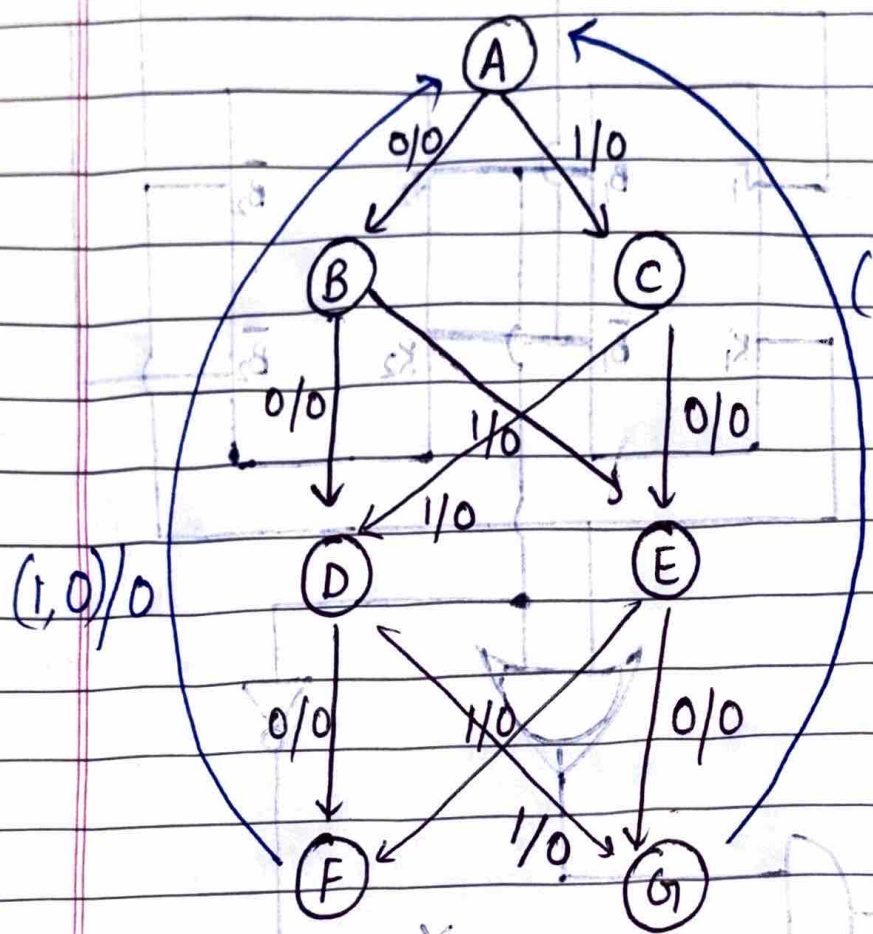
	$\bar{B}_1\bar{B}_0$	$\bar{B}_1 B_0$	$B_1 B_0$	$B_1 \bar{B}_0$
	X	X	0	X
	X	1	0	X
$x B_2$	X	1	1	X
$x \bar{B}_2$	X	X	1	X

$K_0 = \bar{B}_1 + X$

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→ even parity generator



Slight difference in output of last state

Ily solve further.